Laplace transforms of step functions

In the last lesson, we looked at the Laplace transform of $f(t-c)u(t-c)$, the product of $f(t)$ shifted to the right along the t-axis by c units, and a unit step function that "turns on" at $t = c$.

$$\mathscr{L}(f(t-c)u(t-c)) = e^{-cs}F(s)$$

This was the Second Shifting Theorem. Now we want to talk more about the Laplace transforms of step functions, starting with a couple of extensions of the Second Shifting Theorem.

Extending the SST

There are two natural extensions of the Second Shifting Theorem that are worth mentioning. First, if $f(t) = 1$, then the Laplace transform $F(s)$ of $f(t) = 1$ is $F(s) = 1/s$, and the Second Shifting Theorem simplifies to

$$\mathscr{L}(f(t-c)u(t-c)) = e^{-cs}F(s)$$

$$\mathscr{L}(u(t-c)) = e^{-cs}\left(\frac{1}{s}\right)$$

$$\mathscr{L}(u(t-c)) = \frac{e^{-cs}}{s}$$

Of course, because $u_c(t)$ is alternate notation for $u(t-c)$, we can also write this formula as

$$\mathscr{L}(u_c(t)) = \frac{e^{-cs}}{s}$$

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This is an important formula that we'll use regularly to apply the Laplace transform to step functions.

Second, the inverse of the Second Shifting Theorem is also true.

$$\mathscr{L}^{-1}(e^{-cs}F(s)) = f(t-c)u(t-c)$$

Let's do an example where we work backwards to find the original function using the inverse transform.

Example

Find the inverse Laplace transform.

$$G(s) = \frac{se^{-3s}}{(4s-1)(s+2)}$$

We'll start by pulling e^{-3s} out of the fraction, so that the function is then in the form $G(s) = e^{-3s}F(s)$.

$$G(s) = e^{-3s}\frac{s}{(4s-1)(s+2)}$$

Use partial fractions to decompose the remaining fraction, which we'll call $F(s)$.

$$\frac{s}{(4s-1)(s+2)} = \frac{A}{4s-1} + \frac{B}{s+2}$$

To find the value of A, we'll remove the $(4s-1)$ factor, set $s = 1/4$, and then find the value of the fraction on the left.

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$$\frac{s}{s+2}$$

$$\frac{\frac{1}{4}}{\frac{1}{4}+2}=\frac{\frac{1}{4}}{\frac{9}{4}}=\frac{1}{4}\left(\frac{4}{9}\right)=\frac{1}{9}$$

To find the value of B, we'll remove the $(s+2)$ factor, set $s=-2$, and then find the value of the fraction on the left.

$$\frac{s}{4s-1}$$

$$\frac{-2}{4(-2)-1}=\frac{-2}{-8-1}=\frac{-2}{-9}=\frac{2}{9}$$

So the partial fractions decomposition is

$$F(s)=\frac{\frac{1}{9}}{4s-1}+\frac{\frac{2}{9}}{s+2}$$

$$F(s)=\frac{\frac{1}{9}}{4\left(s-\frac{1}{4}\right)}+\frac{\frac{2}{9}}{s+2}$$

$$F(s)=\frac{1}{36}\left(\frac{1}{s-\frac{1}{4}}\right)+\frac{2}{9}\left(\frac{1}{s-(-2)}\right)$$

The inverse transform of $F(s)$ is

$$f(t)=\frac{1}{36}e^{\frac{1}{4}t}+\frac{2}{9}e^{-2t}$$

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Which means the inverse transform of $G(s)$ is

$$\mathscr{L}^{-1}(G(s)) = u(t-3)f(t-3) \text{ with } f(t) = \frac{1}{36}e^{\frac{1}{4}t} + \frac{2}{9}e^{-2t}$$

Fixing the shift

The Second Shifting Theorem and its extensions are all for functions perfectly shifted to $t = c$, $f(t-c)$. But we're often confronted with the Laplace transform of the product of a function and a unit step function, where the function itself isn't shifted by the same distance between 0 and the point where the function "turns on" at $t = c$.

A function like this that doesn't have the right shift can be written as $g(t)u(t-c)$. To find the Laplace transform of $g(t)u(t-c)$, we'll need to rewrite g with the correct shift c.

Essentially, we do this by finding the equation of the curve which is identical to $g(t)$, but shifted c units to the *left*. Then we shift that curve back c units to the *right* to get a curve that's identical to $g(t)$, just rewritten with the shift that we need.

For instance, consider the function $t^3u(t-4)$. The unit step function shows us that we have a 4-unit shift to the right, so we need to rewrite $g(t) = t^3$ with a 4-unit shift to the right as well. But in order to include a shift to the right, while still keeping the same $g(t) = t^3$ curve, we need to start with a curve that's 4 units to the left of $g(t)$, and then shift that curve 4 units to the right, in order to get back to $g(t)$ itself.

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The curve that's 4 units to the left of $g(t) = t^3$, which we'll call $h(t)$, is

$$h(t) = g(t+4) = (t+4)^3$$

$$h(t) = g(t+4) = t^3 + 12t^2 + 48t + 64$$

Now if we shift this curve back 4 units to the right,

$$h(t-4) = g(t+4-4) = (t-4)^3 + 12(t-4)^2 + 48(t-4) + 64$$

$$h(t-4) = g(t) = (t-4)^3 + 12(t-4)^2 + 48(t-4) + 64$$

$$g(t) = (t-4)^3 + 12(t-4)^2 + 48(t-4) + 64$$

we can see that we have a new equation for $g(t)$. This new $g(t)$ is identical to $g(t) = t^3$ (we can graph both of them to prove to ourselves that this is true), it's just been rewritten in terms of a 4-unit shift to the right.

Therefore, we can now rewrite $t^3u(t-4)$ as

$$((t-4)^3 + 12(t-4)^2 + 48(t-4) + 64)u(t-4)$$

$$(t-4)^3u(t-4) + 12(t-4)^2u(t-4) + 48(t-4)u(t-4) + 64u(t-4)$$

And now we've transformed $t^3u(t-4)$ into a form to which we can apply the Second Shifting Theorem $\mathscr{L}(f(t-c)u(t-c)) = e^{-cs}F(s)$, which means we've rewritten $t^3u(t-4)$ as something to which we can apply the Laplace Transform.

$$\mathscr{L}(t^3u(t-4))$$

$$\mathscr{L}((t-4)^3u(t-4) + 12(t-4)^2u(t-4) + 48(t-4)u(t-4) + 64u(t-4))$$

$$\mathscr{L}((t-4)^3u(t-4)) + \mathscr{L}(12(t-4)^2u(t-4)) + \mathscr{L}(48(t-4)u(t-4)) + \mathscr{L}(64u(t-4))$$

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We know from the Second Shifting Theorem that this Laplace transform simplifies to

$$\frac{3!}{s^{3+1}}e^{-4s}+12\left(\frac{2!}{s^{2+1}}\right)e^{-4s}+48\left(\frac{1!}{s^{1+1}}\right)e^{-4s}+64\left(\frac{1}{s}\right)e^{-4s}$$

$$\frac{6}{s^4}e^{-4s}+\frac{24}{s^3}e^{-4s}+\frac{48}{s^2}e^{-4s}+\frac{64}{s}e^{-4s}$$

Even though we were able to get to this transformed result, the algebra required to shift the function to the left and then back to the right definitely got a little complicated. Luckily, there's a formula we can use to fix the shift while avoiding doing all this algebra by hand.

$$\mathscr{L}(g(t)u(t-c))=e^{-cs}\mathscr{L}(g(t+c))$$

In other words, starting over again with $t^3u(t-4)$, using this formula to find the Laplace transform will give us

$$\mathscr{L}(t^3u(t-4))=e^{-4s}\mathscr{L}(g(t+4))$$

$$\mathscr{L}(t^3u(t-4))=e^{-4s}\mathscr{L}((t+4)^3)$$

$$\mathscr{L}(t^3u(t-4))=e^{-4s}\mathscr{L}(t^3+12t^2+48t+64)$$

$$\mathscr{L}(t^3u(t-4))=e^{-4s}\mathscr{L}(t^3)+12e^{-4s}\mathscr{L}(t^2)+48e^{-4s}\mathscr{L}(t)+64e^{-4s}\mathscr{L}(1)$$

and then we can apply transform formulas.

$$\mathscr{L}(t^3u(t-4))=e^{-4s}\left(\frac{3!}{s^{3+1}}\right)+12e^{-4s}\left(\frac{2!}{s^{2+1}}\right)+48e^{-4s}\left(\frac{1!}{s^{1+1}}\right)+64e^{-4s}\left(\frac{1}{s}\right)$$

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$$\mathscr{L}(t^3u(t-4)) = e^{-4s}\left(\frac{6}{s^4}\right) + 12e^{-4s}\left(\frac{2}{s^3}\right) + 48e^{-4s}\left(\frac{1}{s^2}\right) + 64e^{-4s}\left(\frac{1}{s}\right)$$

$$\mathscr{L}(t^3u(t-4)) = \frac{6}{s^4}e^{-4s} + \frac{24}{s^3}e^{-4s} + \frac{48}{s^2}e^{-4s} + \frac{64}{s}e^{-4s}$$

Notice how we get to the same transform using both methods (fixing the shift ourselves algebraically, and using the formula that fixes the shift for us).

Let's do an example where we walk through how to verify that our functions are shifted correctly.

Example

Find the Laplace transform of the function $g(t)$.

$$g(t) = -3u_2(t) + 6(t-4)^2u_4(t) - 9e^{12-2t}u_6(t)$$

We know that the Laplace transform of $u_c(t)$ is given by

$$\mathscr{L}(u_c(t)) = \frac{e^{-sc}}{s}$$

Therefore, the Laplace transform of $-3u_2(t)$ must be

$$-\frac{3e^{-2s}}{s}$$

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To find the transform of $6(t-4)^2u_4(t)$, we identify that $c=4$, and then we can make a substitution of $t+4$ into $f(t-c)$ to determine the function before it was shifted.

$$6(t-4)^2$$

$$6(t+4-4)^2$$

$$6t^2$$

So the Laplace transform will be given by

$$\mathscr{L}(u_c(t)f(t-c)) = e^{-sc}F(s)$$

$$\mathscr{L}(6(t-4)^2u_4(t)) = e^{-4s}\left(6\frac{2!}{s^{2+1}}\right)$$

$$\mathscr{L}(6(t-4)^2u_4(t)) = 6e^{-4s}\left(\frac{2}{s^3}\right)$$

$$\mathscr{L}(6(t-4)^2u_4(t)) = \frac{12e^{-4s}}{s^3}$$

To find the transform of $-9e^{12-2t}u_6(t)$, we identify that $c=6$, and then we can make a substitution of $t+6$ into $f(t-c)$ to determine the function before it was shifted.

$$-9e^{12-2t}$$

$$-9e^{12-2(t+6)}$$

$$-9e^{12-2t-12}$$

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$$-9e^{-2t}$$

So the Laplace transform will be given by

$$\mathscr{L}(u_c(t)f(t-c)) = e^{-sc}F(s)$$

$$\mathscr{L}(-9e^{12-2t}u_6(t)) = e^{-6s}\left(-9\frac{1}{s-(-2)}\right)$$

$$\mathscr{L}(-9e^{12-2t}u_6(t)) = -\frac{9e^{-6s}}{s+2}$$

Putting these three transforms together, the Laplace transform is

$$\mathscr{L}(g(t)) = -\frac{3e^{-2s}}{s} + \frac{12e^{-4s}}{s^3} - \frac{9e^{-6s}}{s+2}$$

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